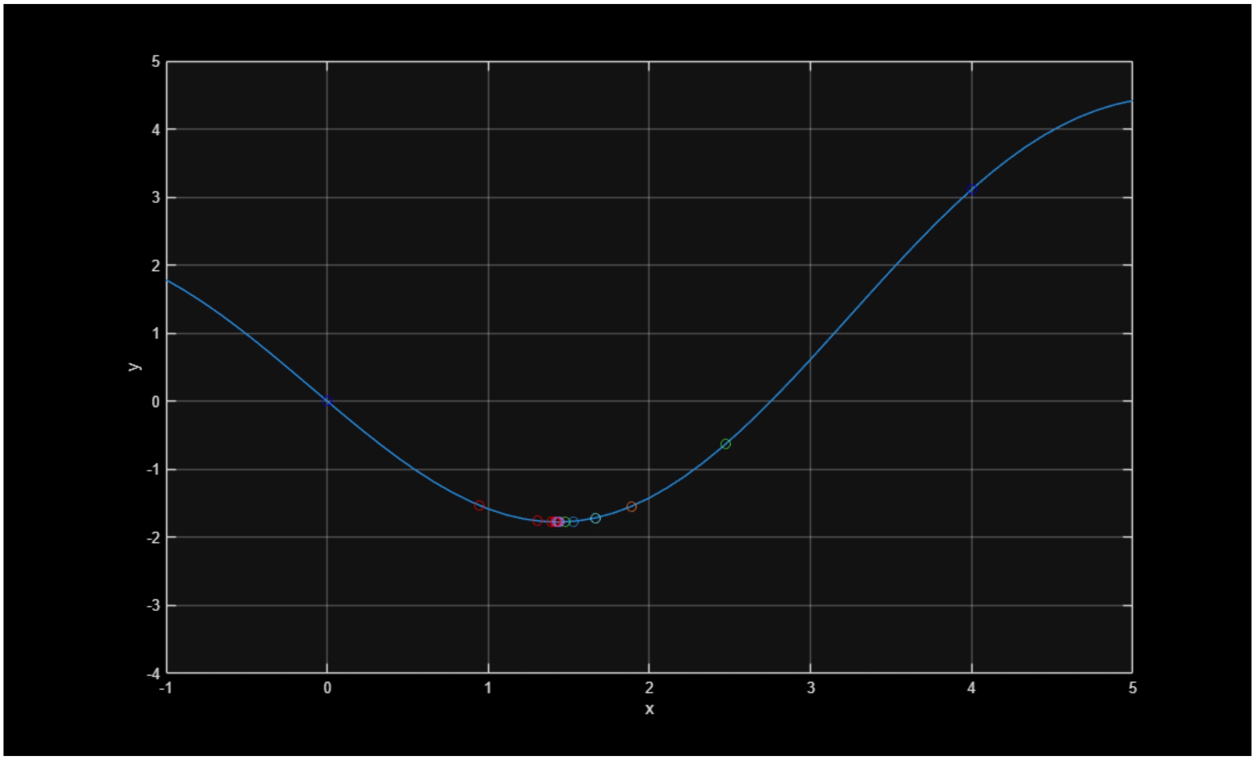
```
% Inclass 3

    clear; clc; close all;

    f_x =@(x) x.^2/10 - 2*sin(x);

    x = linspace(-1,10,100);
    y = f_x(x);

    plot(x,y,'-')
    axis([-1 5 -4 5])
    xlabel('x')
    ylabel('y')
    grid on
    hold on

    xL = 0; xU =4;
    fL = f_x(xL);   fU = f_x(xU);
    plot(xL,fL,'bo')
    plot(xU,fU,'bo')

    phi = (1+sqrt(5))/2;
    iter = 0;
    while (1)
        iter = iter + 1;

        d = (phi-1)*(xU - xL);

        x1 = xL + d;    f1 = f_x(x1);
        x2 = xU - d;    f2 = f_x(x2);

        if f1 < f2
            x_opt = x1;
            xL = x2;
            fL = f2;    f_opt = f1;
        plot(xL,fL,'ro')
        plot(x_opt,f_opt,'ro')
        else
            x_opt = x2;
            xU = x1;
            fU = f1;    f_opt = f2;
        plot(x_opt,f_opt,'o')
        plot(xU,fU,'o')
pause
        end

            err = (2-phi)*abs((xU-xL)/x_opt)*100;

        fprintf('%5d %9.2f %9.2f %9.2f %9.6f \n',iter, xL, x_opt,xU,err);

            if err < 1e-6 || iter > 50, break, end
```

```
end
hold off
```

```
 1      0.00       1.53       2.47 61.803399
 2      0.94       1.53       2.47 38.196601
 3      0.94       1.53       1.89 23.606798
 4      1.30       1.53       1.89 14.589803
 5      1.30       1.53       1.67  9.016994
 6      1.30       1.44       1.53  5.901699
 7      1.39       1.44       1.53  3.647451
 8      1.39       1.44       1.48  2.254249
 9      1.39       1.42       1.44  1.412887
10      1.41       1.42       1.44  0.873212
11      1.42       1.43       1.44  0.536778
12      1.42       1.43       1.44  0.331747
13      1.42       1.43       1.43  0.205452
14      1.43       1.43       1.43  0.126976
15      1.43       1.43       1.43  0.078476
16      1.43       1.43       1.43  0.048501
17      1.43       1.43       1.43  0.029975
18      1.43       1.43       1.43  0.018526
19      1.43       1.43       1.43  0.011448
20      1.43       1.43       1.43  0.007075
21      1.43       1.43       1.43  0.004373
22      1.43       1.43       1.43  0.002702
23      1.43       1.43       1.43  0.001670
24      1.43       1.43       1.43  0.001032
25      1.43       1.43       1.43  0.000638
26      1.43       1.43       1.43  0.000394
27      1.43       1.43       1.43  0.000244
28      1.43       1.43       1.43  0.000151
29      1.43       1.43       1.43  0.000093
30      1.43       1.43       1.43  0.000058
31      1.43       1.43       1.43  0.000036
32      1.43       1.43       1.43  0.000022
33      1.43       1.43       1.43  0.000014
34      1.43       1.43       1.43  0.000008
35      1.43       1.43       1.43  0.000005
36      1.43       1.43       1.43  0.000003
37      1.43       1.43       1.43  0.000002
38      1.43       1.43       1.43  0.000001
39      1.43       1.43       1.43  0.000001
```



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